

Lab 1 (CS201)

1. Accuracy of Taylor expansion: Develop a Taylor expansion for the function $f(x) = \frac{1}{x+1}$. Plot the function between $x = 0$ and $x = 1$. Compute a Taylor series expansion of the function around $a = 0, 0.5, 0.8$ and for number of terms $n = 3, 5$. In each case, make a plot of the function $f(x)$ and the Taylor approximation. Are the results as expected. Justify your answer.

Remark: This problem is meant to highlight the effect the number of terms included in the Taylor approximation have on the accuracy as well as the influence of the point around which the expansion is performed.

2. Taylor series of an integral: Consider the function,

$$f(x) = \frac{1}{x} \int_0^x \frac{\sin(t)}{t}, \text{ where } x \neq 0$$

Develop a Taylor polynomial approximation with an error equal to or less than 5×10^{-9} on the interval $|x| < 1$. Plot the error over the entire range and show that it always remains lower than the prescribed bound. Note that the function is a known function known as the sine integral and is generally included in scipy and Matlab, enabling comparison with the approximation.

3. Particle falling through viscous medium Consider a particle falling vertically under gravity in a viscous medium. Assume the drag force is modeled by, $F_{\text{drag}} = -bv$ where b is a constant and v is the velocity of the particle. Set $b/m = 2.0s^{-1}$ and acceleration due to gravity $g = 10m/s$ and plot the position and the velocity of the particle against time, beginning at $t = 0s$ and ending at $t = 3.0s$. Utilise Euler's method for this purpose, with the following initial conditions, $x(0) = 10m, v(0) = 0m/s$. Do this for two case:
 - (a) The number of subintervals is $n = 50$ and,
 - (b) The number of subintervals is $n = 500$.

In each case, compare with the known analytical solution and comment on the accuracy.

Lastly, set b equal to zero and compute the position and the velocity with $n = 500$. Comment on the difference between $b = 0$ and $b \neq 0$.